

§2.4 Partitioned matrices

Recall we have seen we can partition a matrix into cols.
More ways to partition a matrix. It will be sometimes
useful for applications and calculations.

Exmp 1

$$A = \begin{bmatrix} 3 & 0 & -1 & | & 5 & 9 & | & -2 \\ -5 & 2 & 4 & | & 0 & -3 & | & 1 \\ -8 & -6 & 3 & | & 1 & 7 & | & -4 \end{bmatrix}$$

A_{11} A_{12} A_{13} A_{21} A_{22} A_{23}

can be written as the 2×3 partitioned or block matrix

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}$$

notice that A_{ij} 's are of different sizes.

Addition & scalar mult.

A, B same size, are partitioned in the same way.

$A + B$: add the corresponding blocks

λA : multiply each block by λ .

Multiplication of partitioned matrices

AB

Imitate the row-column rule, provided the column partition of A matches the row partition of B (so that all multiplications make sense)

Exmp 3

$$A = \begin{bmatrix} 2 & -3 & 1 & | & 0 & -4 \\ 1 & 5 & -2 & | & 3 & -1 \\ 0 & -4 & -2 & | & 7 & -1 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

$$B = \begin{bmatrix} 6 & 4 \\ -2 & 1 \\ -3 & 7 \\ -1 & 3 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

We say partitions of A and B are conformable for block multi.

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} A_{11}B_1 + A_{12}B_2 \\ A_{21}B_1 + A_{22}B_2 \end{bmatrix}$$

Warning: order.

Multiplications make sense.

Column-Row expansion of AB

Exmp $A = \begin{bmatrix} -3 & 1 & 2 \\ 1 & -4 & 5 \end{bmatrix}$ $B = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$

$$AB = \begin{bmatrix} -3a + c + 2e & -3b + d + 2f \\ a - 4c + 5e & b - 4d + 5f \end{bmatrix}$$

$$\text{col}_1(A) \cdot \text{row}_1(B) = \begin{bmatrix} -3 \\ 1 \end{bmatrix} \begin{bmatrix} a & b \end{bmatrix} = \begin{bmatrix} -3a & -3b \\ a & b \end{bmatrix}$$

$$\text{col}_2(A) \cdot \text{row}_2(B) = \begin{bmatrix} 1 \\ -4 \end{bmatrix} \begin{bmatrix} c & d \end{bmatrix} = \begin{bmatrix} c & d \\ -4c & -4d \end{bmatrix}$$

$$\text{col}_3(A) \cdot \text{row}_3(B) = \begin{bmatrix} 2 \\ 5 \end{bmatrix} \begin{bmatrix} e & f \end{bmatrix} = \begin{bmatrix} 2e & 2f \\ 5e & 5f \end{bmatrix}$$

Thus, $AB = \text{col}_1(A) \text{row}_1(B) + \text{col}_2(A) \text{row}_2(B) + \text{col}_3(A) \text{row}_3(B)$.

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More generally,

Thm 10 $A: m \times n$ matrix $B: n \times p$ matrix. Then

$$AB = \text{col}_1(A) \text{row}_1(B) + \text{col}_2(A) \text{row}_2(B) + \dots + \text{col}_n(A) \text{row}_n(B)$$

Inverses

Exmp $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ invertible. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} a^{-1} & -\frac{b}{ad-bc} \\ -\frac{c}{ad-bc} & d^{-1} \end{bmatrix}$

Q How about a block matrix?

Exmp $A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$ Assume $A_{11} = p \times p$ mat. $A_{22} = q \times q$ mat. and A invertible.

Find A^{-1} .

Sol Let $B = A^{-1}$. Then $AB = I_{p+q}$

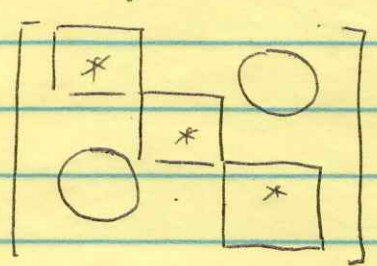
We partition $B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$ s.t. $\begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} I_p & 0 \\ 0 & I_q \end{bmatrix}$

Then, (LHS) = $\begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{22}B_{21} & A_{22}B_{22} \end{bmatrix} = \begin{bmatrix} I_p & 0 \\ 0 & I_q \end{bmatrix}$

Hence,

$$\begin{cases} A_{22} B_{22} = I_q \\ A_{22} B_{21} = 0 \\ A_{11} B_{11} + A_{12} B_{21} = I_p \\ A_{11} B_{12} + A_{12} B_{22} = 0 \end{cases} \Rightarrow \begin{cases} B_{22} = A_{22}^{-1} \\ B_{21} = A_{22}^{-1} 0 = 0 \\ B_{11} = A_{11}^{-1} \\ B_{12} = -A_{11}^{-1} A_{12} A_{22}^{-1} \end{cases}$$

• A block diagonal matrix:



zero blocks off the main diagonal.

$I \Leftrightarrow$ invertible iff each block on the diagonal is invertible.